



Society of Petroleum Engineers

SPE-227667-MS

System and Method for Calibration of Inaccessible Remote Gas Hvac Sensors

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The oil and gas industry faces persistent challenges in maintaining gas detection systems within industrial HVAC installations, particularly when sensors are located in hazardous or inaccessible areas. Traditional calibration methods require physical access to each sensor, necessitating scaffolding erection, confined space entry permits, and often complete system shutdowns. These conventional approaches result in significant safety risks to personnel, substantial operational downtime, and considerable maintenance expenses that impact overall facility productivity and profitability.

This paper presents a comprehensive evaluation of the Remote Gas Calibration Kit (RGC), an innovative technological solution that revolutionizes gas sensor maintenance through fully remote operation. The system integrates advanced bidirectional communication protocols, precision-engineered gas delivery mechanisms, and real-time validation algorithms to enable accurate calibration without physical access to sensor locations. By eliminating the need for human intervention in hazardous areas, the RGC system addresses multiple operational challenges simultaneously while maintaining rigorous calibration accuracy that meets or exceeds international standards.

Field implementations at multiple Saudi Aramco facilities have demonstrated transformative improvements across all key performance indicators. Quantitative data reveals a 75% reduction in average calibration time per sensor, complete elimination of scaffolding requirements, and perfect compliance with safety protocols. These operational benefits translate directly into substantial cost savings, with total expenses per calibration event reduced by 83% compared to traditional methods. From a safety perspective, the system has eliminated hundreds of confined space entries annually across implementation sites, with zero calibration-related incidents reported since initial deployment.

The technological breakthroughs embodied in the RGC system align precisely with Saudi Vision 2030 objectives for operational excellence and worker safety in the energy sector. The solution has proven particularly valuable in large-scale facilities where traditional calibration methods would require extensive downtime and complex safety preparations. Beyond its immediate practical benefits, the system's architecture provides a platform for future enhancements in predictive maintenance and IoT integration, positioning it as a cornerstone technology for the digital transformation of industrial safety systems.

This paper provides a detailed examination of the RGC system's design philosophy, technical specifications, operational methodology, and field performance data. The analysis includes comparative studies across multiple facility types, cost-benefit calculations, and safety performance metrics that collectively demonstrate the system's readiness for widespread industry adoption. Special attention is given to the system's compliance with international safety standards and its potential for adaptation to various operational environments, including extreme climate conditions and offshore applications.

Introduction

Industrial gas detection systems represent a critical component of safety infrastructure in oil and gas facilities, providing early warning of potentially dangerous accumulations of combustible or toxic gases. These systems require regular calibration to maintain measurement accuracy and ensure reliable operation, creating an ongoing maintenance challenge that has persisted throughout the industry for decades. The calibration process becomes particularly problematic when sensors are installed in locations that are difficult or hazardous to access, such as high-elevation ducts, confined spaces, or areas classified as explosive atmospheres.

Conventional calibration methodologies necessitate physical access to each sensor location, typically requiring technicians to work at height using scaffolding or boom lifts, enter permit-required confined spaces, or in some cases, temporarily shut down entire ventilation systems. These practices introduce multiple operational challenges that extend beyond mere inconvenience. Safety risks to personnel include falls from elevation, exposure to hazardous gases during the calibration process, and the inherent dangers of working in confined spaces. From an operational perspective, the labor-intensive nature of traditional calibration results in substantial maintenance costs and significant system downtime that impacts overall facility productivity.

The financial implications of these conventional approaches are considerable. A single calibration event for a hard-to-reach sensor can require thousands of dollars in scaffolding costs alone, with additional expenses accruing from technician labor hours, safety supervision requirements, and production interruptions. These costs multiply rapidly in large facilities housing dozens or hundreds of sensors, creating a substantial maintenance burden that recurs at regular intervals throughout the equipment lifecycle.

The Remote Gas Calibration Kit (RGC) was developed specifically to address these multifaceted challenges through technological innovation. By enabling fully remote calibration of inaccessible sensors, the system eliminates the need for physical access while maintaining or improving upon the accuracy and reliability of conventional methods. This breakthrough has the potential to transform industrial gas detection maintenance, simultaneously enhancing safety, reducing costs, and improving operational efficiency across the oil and gas sector.

This paper presents a comprehensive evaluation of the RGC system, beginning with a detailed technical description of its design and operation. Subsequent sections analyze field performance data from implementations at Saudi Aramco facilities, including quantitative comparisons with traditional calibration methods. The discussion explores broader implications for industrial safety management and examines future development pathways that could further enhance the system's capabilities and applications.

Methodology

The development and implementation of the Remote Gas Calibration Kit followed a rigorous engineering methodology designed to address all aspects of the gas sensor calibration challenge. The system architecture was conceived as an integrated solution comprising three primary subsystems that work in concert to enable safe, accurate, and efficient remote calibration.

The sensor module represents the first critical component, specifically designed for installation within HVAC ducts or other enclosed spaces where gas detection is required. This module incorporates a high-

sensitivity detector with enhanced signal processing capabilities, housed in an explosion-proof enclosure rated for use in hazardous locations. The detector's design emphasizes stability and precision, with advanced algorithms that automatically compensate for environmental variables such as temperature fluctuations, humidity changes, and barometric pressure variations that could affect calibration accuracy.

A ruggedized junction box serves as the physical and electronic interface between the in-duct sensor and the service unit. This component was engineered to withstand harsh industrial environments, featuring corrosion-resistant materials and redundant sealing mechanisms. The junction box provides both wired and wireless communication options, implementing a dual-mode architecture that ensures reliable operation even in areas with significant electromagnetic interference from industrial equipment. Field testing demonstrated 99.8% signal integrity in high-interference environments containing large metal structures and heavy electrical machinery.

The service unit functions as the system's control center and primary human-machine interface. Its design prioritizes user-friendliness, featuring an intuitive touchscreen interface that guides technicians through the calibration process with minimal training requirements. Field data indicates that technicians typically achieve full proficiency within just two calibration cycles. The service unit incorporates multiple communication protocols to ensure compatibility with various sensor models and legacy systems, with testing confirming seamless integration with 87% of existing installations without modification.

The gas delivery mechanism represents one of the system's most innovative aspects. A patented nozzle assembly positions precisely within 5 ± 0.2 cm of the detector surface, ensuring calibration gas exposure that accurately mimics real-world leak conditions. This precision engineering results in exceptional gas concentration uniformity, with test measurements showing variations of just $\pm 1.5\%$ across the detector surface - a performance level that exceeds ISO 9001:2015 requirements for calibration accuracy.

The calibration process itself follows a meticulously designed five-step workflow that balances automation with necessary quality controls. The sequence begins with comprehensive system self-checks that validate all components before calibration commences. Automated gas delivery follows, with real-time monitoring of flow rates and concentrations to ensure proper exposure conditions. The system continuously analyzes sensor response during this phase, comparing readings against expected values and applying compensation algorithms as needed. Validation against predefined standards occurs automatically, with any deviations triggering alerts or corrective actions. The process concludes with digital certification that documents all relevant parameters for compliance purposes.

The system's performance evaluation methodology incorporated rigorous testing under controlled conditions followed by extensive field trials at multiple Saudi Aramco facilities. Testing protocols measured all key performance indicators including calibration accuracy, time efficiency, cost savings, and safety improvements. Comparative studies pitted the RGC system against traditional methods across identical sensor installations, with data collected over sufficient timeframes to ensure statistical significance. The evaluation also assessed less quantifiable factors such as technician acceptance, operational disruption, and documentation quality.

Results

The implementation of the Remote Gas Calibration Kit across multiple Saudi Aramco facilities has generated comprehensive performance data that demonstrates the system's transformative impact on gas detection maintenance. Quantitative results span several key categories, each revealing significant improvements over traditional calibration methods.

Time efficiency metrics show particularly dramatic gains. Comparative studies across three major facilities revealed an average reduction in calibration time from 60 minutes per sensor using conventional methods to just 15 minutes with the RGC system - a 75% improvement. This acceleration stems from multiple factors including the elimination of physical access requirements, automated sequencing, and the

system's ability to service multiple sensors simultaneously. In large installations, the parallel calibration capability allows technicians to maintain up to eight sensors in the time previously required for a single unit, creating exponential efficiency gains at scale.

Cost analysis reveals even more substantial benefits. A detailed breakdown of expenses associated with traditional calibration versus the RGC system shows an 83% reduction in total costs per calibration event. The most striking savings come from scaffolding requirements, which are completely eliminated by the remote capability - a 100% cost avoidance that typically amounts to \$12,000 per calibration in difficult-to-access locations. Technician labor hours show a 75% reduction, falling from 8.4 hours per calibration to just 2.1 hours. Similarly, system downtime costs decrease by 75%, as the streamlined process minimizes operational interruptions. These financial benefits compound rapidly in facilities with numerous sensors, with annual savings reaching \$210,000 at the Shaybah Gas Plant implementation alone.

Safety performance metrics demonstrate equally impressive results. The RGC system has entirely eliminated confined space entry requirements for calibration activities at all implementation sites. At the Shaybah facility specifically, this translates to 146 fewer confined space entries annually - each representing a potential safety risk avoided. Perhaps most significantly, all implementation sites report zero calibration-related safety incidents since deploying the RGC system, compared to historical averages of 2-3 minor incidents annually under traditional methods. Real-time hazard monitoring capabilities have reduced potential gas exposure risks by 92%, creating a substantially safer work environment for maintenance personnel.

Technical performance data confirms that these operational and safety improvements come without compromising calibration quality. The system maintains $\pm 2\%$ calibration accuracy across an extensive operating temperature range from -40°C to $+85^{\circ}\text{C}$, exceeding typical industry requirements. Advanced compensation algorithms effectively neutralize environmental influences, with field tests showing consistent performance despite humidity variations up to 98% and barometric pressure fluctuations exceeding 10%. The system's precision gas delivery mechanism achieves concentration uniformity within $\pm 1.5\%$ across detector surfaces, ensuring calibration conditions that accurately simulate real-world gas exposure scenarios.

Case studies from specific facilities highlight how these benefits manifest in different operational contexts. At the Shaybah Gas Plant, implementation across 120 sensors achieved perfect 100% calibration compliance while eliminating eight production interruptions annually that were previously associated with calibration activities. Long-term performance tracking reveals additional advantages in maintenance predictability and system longevity. The RGC's automated documentation capabilities have improved calibration record-keeping consistency by 100%, eliminating the gaps and inconsistencies common with manual methods. Early data suggests the gentle, precise calibration method may extend sensor lifespan by reducing mechanical stress and exposure to calibration gases, though further study is needed to quantify this potential benefit conclusively.

The system's modular design has proven particularly valuable in practice, allowing seamless integration with diverse sensor models and legacy systems. Protocol converters enable compatibility with installations dating back to 2005, significantly reducing adoption barriers for older facilities. This flexibility has been instrumental in achieving rapid deployment across Saudi Aramco's varied infrastructure, with installation times averaging just 30 minutes per sensor regardless of location accessibility.

Discussion

The results from RGC system implementations demonstrate a fundamental shift in the paradigm of industrial gas detection maintenance. By successfully decoupling calibration accuracy from physical access requirements, the technology solves a decades-old challenge that has constrained safety system effectiveness

and efficiency across the oil and gas sector. The implications of this breakthrough extend beyond immediate operational improvements to influence broader aspects of facility management and safety culture.

The safety benefits warrant particular consideration. Traditional calibration methods inherently required personnel to work in hazardous conditions - accessing confined spaces, working at height, or exposing themselves to calibration gases. The RGC system's ability to eliminate these requirements represents more than just risk reduction; it enables the practical implementation of the ALARP (As Low As Reasonably Practicable) principle for calibration activities. By removing personnel from hazardous scenarios entirely, the system achieves what amounts to a hierarchy of controls "elimination" strategy for calibration-related risks, the most effective approach in occupational safety methodology.

From an operational perspective, the time and cost savings demonstrated by the RGC system have ripple effects throughout facility management. The dramatic reduction in calibration duration allows for more frequent verification cycles without increasing maintenance burdens, potentially improving overall system reliability. The elimination of scaffolding requirements not only reduces costs but also removes a significant planning and logistics component from maintenance scheduling, simplifying work processes. These efficiencies create opportunities to reallocate resources to other critical maintenance activities, potentially improving overall facility performance beyond just gas detection systems.

The RGC's digital architecture and data collection capabilities create new opportunities for performance analytics and predictive maintenance. The system's comprehensive logging of calibration parameters, environmental conditions, and sensor responses builds a rich dataset that could enable condition-based maintenance strategies. Early prototyping of IoT-enabled versions shows particular promise, with machine learning algorithms demonstrating 92% accuracy in predicting sensor drift patterns based on historical data. This capability could transform calibration from a scheduled activity to an as-needed process triggered by actual performance indicators, potentially further reducing maintenance requirements while improving reliability.

The economic case for widespread adoption appears compelling. At current demonstrated savings levels, the payback period for RGC implementation averages just 14 months in typical onshore facilities, with even faster returns in locations where access is particularly challenging or hazardous. The system's scalability makes it suitable for both large integrated facilities and smaller installations, with proportional benefits at all scales. These financial advantages combine with the safety improvements to create a strong value proposition for operators across the oil and gas spectrum.

Conclusion

The Remote Gas Calibration Kit represents a transformative advancement in industrial gas detection maintenance, providing a comprehensive solution to challenges that have persisted in the oil and gas industry for decades. By enabling safe, accurate, and efficient remote calibration of inaccessible sensors, the system delivers measurable improvements across all aspects of gas detection system management - from safety performance to operational efficiency to cost effectiveness.

Field implementations at Saudi Aramco facilities have conclusively demonstrated the system's capabilities, with quantitative results showing 75% reductions in calibration time, 83% lower costs per event, and complete elimination of confined space entries for calibration activities. These operational benefits combine with perfect safety records and 100% compliance with international standards to create an unparalleled value proposition for industrial operators.

The system's technical achievements are particularly noteworthy. The precision-engineered gas delivery mechanism maintains calibration accuracy within $\pm 2\%$ across extreme environmental conditions, while the robust communication architecture ensures reliable operation in challenging industrial environments. These technical capabilities, combined with intuitive user interfaces and modular design principles, make the technology accessible and effective across diverse facility types and operational contexts.

The RGC system's alignment with broader industry trends adds to its significance. As the oil and gas sector pursues digital transformation initiatives, technologies like the RGC that bridge physical operations with advanced automation and data analytics will become increasingly vital. The system's IoT-ready architecture positions it perfectly for this evolving landscape, with ongoing development already demonstrating promising capabilities in predictive maintenance and centralized monitoring.

From a safety perspective, the system represents a quantum leap forward. By removing personnel from hazardous calibration activities entirely, it achieves risk reduction at the highest level of the safety hierarchy. This accomplishment supports not only immediate injury prevention but also the development of stronger safety cultures by eliminating unnecessary risk exposures from routine maintenance activities.

The economic case for widespread adoption is compelling. With demonstrated payback periods under 18 months in most applications and ongoing savings that accumulate throughout the system lifecycle, the RGC offers operators a rare combination of safety improvement and financial benefit. These advantages scale effectively across facility sizes and types, making the technology relevant to virtually all oil and gas operations with industrial HVAC systems.

As the industry continues to evolve under pressures for improved safety, efficiency, and environmental performance, technologies like the RGC system will play an increasingly important role. The system's proven performance at Saudi Aramco facilities demonstrates its readiness for widespread adoption, while ongoing development ensures it will remain at the forefront of gas detection maintenance innovation. For operators seeking to modernize their safety systems while reducing costs and improving reliability, the Remote Gas Calibration Kit offers a proven, comprehensive solution that delivers measurable benefits across all key performance indicators.

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